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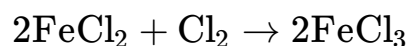
display the question in the output and answer these question of o level chemistry. explain in detail



Here are the four multiple-choice questions from the image, along with detailed answers and explanations suitable for O Level Chemistry.

Question 1

Chlorine reacts with iron(II) chloride.



Which statement about the reaction is correct?

- A. The iron(II) chloride is an oxidizing agent and is oxidized.
- B. The iron(II) chloride is an oxidizing agent and is reduced.
- C. The iron(II) chloride is a reducing agent and is oxidized.
- D. The iron(II) chloride is a reducing agent and is reduced.

✓ Correct answer: C

Explanation (step-by-step):

1. Assign oxidation states:

- In FeCl_2 : Fe = +2, Cl = -1.
- In FeCl_3 : Fe = +3, Cl = -1.
- In Cl_2 : Cl = 0.

2. Identify changes:

- Fe goes from +2 → +3 (loss of electrons → **oxidation**).
- Cl_2 (0) goes to Cl (-1) in FeCl_3 (gain of electrons → **reduction**).

3. Roles:

- **FeCl_2** provides electrons (causes Cl_2 to be reduced) → FeCl_2 is the **reducing agent**.

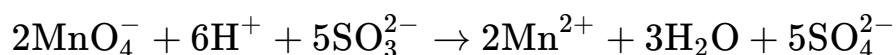
- A reducing agent gets **oxidized** itself.

4. Check options:

- C says: "The iron(II) chloride is a reducing agent and is oxidized."  Correct.

Question 2

Reaction:



Which ion is reduced and what is the reducing agent?

Option	Ion that is reduced	Reducing agent
A	MnO_4^-	H^+
B	MnO_4^-	SO_3^{2-}
C	SO_3^{2-}	H^+
D	SO_3^{2-}	MnO_4^-

 Correct answer: B

Explanation:

1. Oxidation states:

- In MnO_4^- : Mn = +7.
- In Mn^{2+} : Mn = +2 (gain of electrons → **reduction**).
So MnO_4^- is reduced.

2. In SO_3^{2-} : S = +4.

- In SO_4^{2-} : S = +6 (loss of electrons → **oxidation**).
So SO_3^{2-} is oxidized.

3. Reducing agent = substance oxidized = SO_3^{2-} .

4. H^+ stays at +1 throughout, not redox active.

Thus: **Reduced ion = MnO_4^- , Reducing agent = SO_3^{2-}** → Option B.

Question 3

Chemical Z is a powerful reducing agent.

Which statement about Z is correct?

- A. Z reacts with aqueous potassium iodide producing a brown solution and gains electrons in the process.
- B. Z reacts with aqueous potassium iodide producing a brown solution and loses electrons in the process.
- C. Z decolourises acidified potassium manganate(VII) and gains electrons in the process.
- D. Z decolourises acidified potassium manganate(VII) and loses electrons in the process.

✓ Correct answer: **D**

Explanation:

1. **A reducing agent** loses electrons (is oxidized).
2. **Test with acidified KMnO_4 (purple):**
 KMnO_4 is an oxidizing agent.
 A reducing agent will reduce purple MnO_4^- to colourless Mn^{2+} → decolorization.
 In doing so, the reducing agent Z **loses electrons** (is oxidized). → Matches **D**.
3. **Check the KI case (A & B):**
 KI contains I^- (a reducing agent).
 A reducing agent Z would not oxidize I^- to brown I_2 (that requires an oxidizing agent).
 So A and B are wrong because a reducing agent won't turn KI brown.

Question 4

Given:

- Aqueous FeCl_3 reacts with aqueous KI → brown solution.
- Aqueous FeCl_2 reacts with acidified KMnO_4 → yellow/pale orange solution.

What is the role of FeCl_3 and FeCl_2 ?

Option	FeCl_3	FeCl_2
A	oxidising agent	reducing agent
B	reducing agent	oxidising agent
C	oxidising agent	oxidising agent
D	reducing agent	reducing agent

✓ Correct answer: **A**

Explanation:

1. **Reaction 1: $\text{FeCl}_3 + \text{KI} \rightarrow \text{brown solution (I}_2\text{)}$**

- $\text{I}^- \rightarrow \text{I}_2$ (oxidation, loses electrons).
- Fe^{3+} must accept electrons $\rightarrow$ reduced to Fe^{2+} .
- So FeCl_3 is the **oxidizing agent**.

2. Reaction 2: FeCl_2 + acidified $\text{KMnO}_4 \rightarrow$ yellow/pale orange (Fe^{3+} formed)

- KMnO_4 (purple) is reduced to Mn^{2+} (pale pink, but with Fe^{3+} yellow-orange seen).
- $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+}$ (oxidation, loses electrons).
- So FeCl_2 is the **reducing agent**.

Therefore: FeCl_3 = oxidizing agent, FeCl_2 = reducing agent $\rightarrow$ **Option A**.

Final Answers Summary

1. **C**
2. **B**
3. **D**
4. **A**